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Agent Type: Restaurant Booking Agent

1. Identify the Problem

Usually, when someone wants to book a table at a restaurant, they first have to call or go there in person, which takes a lot of time and also makes it difficult to compare different options. To solve this problem, we have designed a Restaurant Agent that takes the user's preferred cuisine or area as input, searches for restaurants accordingly, shows the available options along with their ratings, and books a table easily once the user makes a choice.

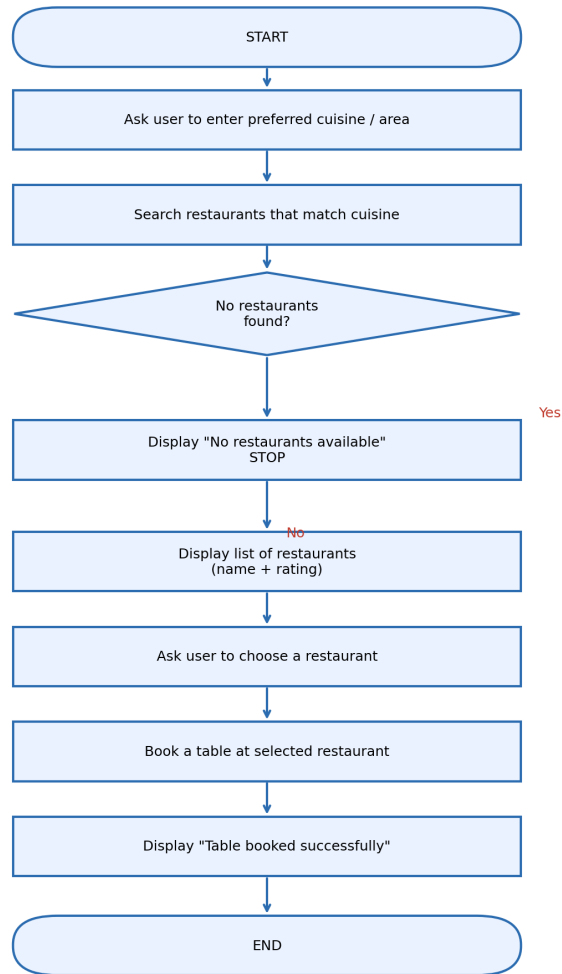
2. Formulation

2.1 Textual Formulation

```
START
    Ask the user to enter preferred cuisine (or area)
    Search for restaurants that match the cuisine

IF no restaurants are found THEN
    Display "No restaurants available"
    STOP
ELSE
    Display list of available restaurants with name and rating
    Ask user to choose a restaurant
    Book a table at the selected restaurant
    Display "Table booked successfully"
END IF
END
```

2.2 Graphical Formulation (Flowchart)



3. Evaluation in the Context of PEAS

Element	Description
Performance Measure	How accurately the agent matches a restaurant to the user's demand, whether the booking completes properly, and how fast it responds.
Environment	The user, the restaurant data with cuisines and ratings, and the availability of tables.
Actuators	Showing the list of matching restaurants on screen, displaying messages, and confirming the booking.
Sensors	The user's input (the requested cuisine or area), and the restaurant selected from the list.

4. Environmental Properties

Property	Explanation
Partially Observable	The agent only knows what the user tells it (the cuisine or area); it does not have a direct view of the whole restaurant environment.
Deterministic	If the user enters the same cuisine or area again, the agent will always return the same list of matching restaurants.
Sequential	All the steps happen in a particular order: first search, then show the results, then book, so actions taken earlier affect the later steps.
Static	The restaurant list does not change while the agent is searching or waiting for the user's response.
Discrete	Cuisines, restaurant names, and ratings all come from a fixed, countable set, not a continuous range.
Single-agent	Only this one restaurant booking agent is working in this environment; there is no other competing agent.